

MIND-BLOOM: Interpretable Machine Learning for Prediction of Postpartum Depression: A Case Study Demonstrating Global Applicability

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Abstract—Postpartum depression (PPD) affects an estimated 10–20% of mothers worldwide, with even higher prevalence consistently reported in low- and middle-income countries (LMICs) where structured postnatal mental health screening remains limited or completely absent [1]. This paper presents MIND-BLOOM, an interpretable machine learning framework for early PPD prediction that is designed to be applicable across diverse health systems, and is demonstrated through a detailed case study using an open Bangladeshi survey dataset (n=800) containing EPDS/PHQ scores and extensive socio-demographic, familial, obstetric, and neonatal features [2]. Building on recent explainable PPD risk models developed in Chinese tertiary hospitals [3]–[5], MIND-BLOOM employs calibrated logistic regression, support vector machines (SVMs), and tree-based ensembles (random forest, XGBoost) together with SHAP explanations, while explicitly addressing LMIC challenges through culturally adapted feature sets, careful handling of class imbalance, and subgroup fairness analyses. The evaluation framework combines stratified training, validation, and testing splits, nested cross-validation, and decision curve analysis to optimize clinically meaningful risk thresholds that remain robust across different health system resource levels. By integrating globally relevant risk factors with locally grounded data, interpretable algorithms, and deployment considerations for low-resource maternal health services, MIND-BLOOM aims to support scalable PPD screening, risk stratification, and referral worldwide, with particular emphasis on LMIC contexts.

Index Terms—Postpartum depression, interpretable machine learning, SHAP, maternal mental health, LMICs, risk prediction.

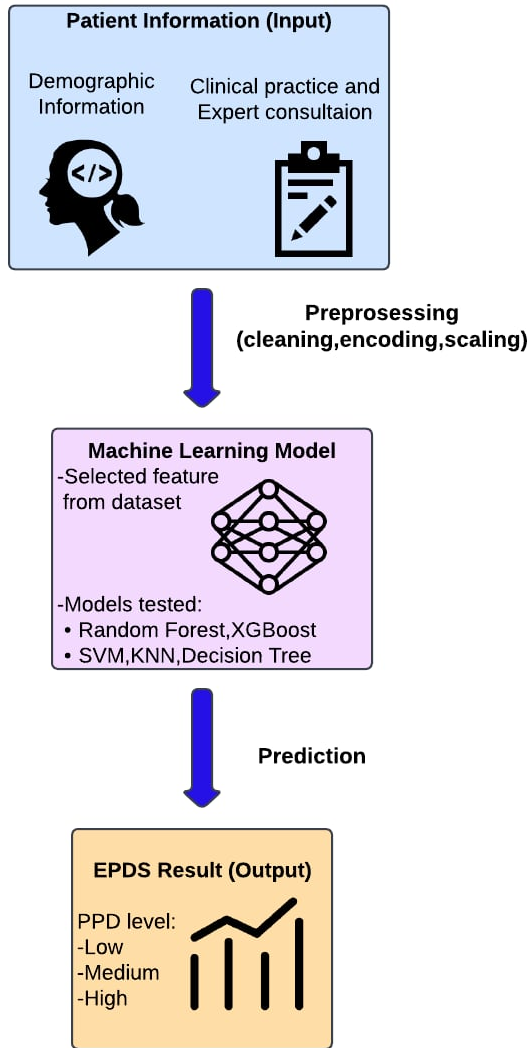
I. INTRODUCTION

Postpartum depression (PPD) is a common and debilitating mental health condition that undermines maternal functioning, infant bonding, and long-term family well-being. Global estimates suggest that 10–20% of mothers experience clinically significant depressive symptoms in the first year after childbirth, with prevalence rising to 47.8–60.6% in several LMIC facility-based studies where social stressors and limited mental health services co-occur [1], [6]–[8]. Despite this substantial burden, routine PPD screening and standardized risk assessment are implemented inconsistently across health systems, and are particularly scarce in resource-limited settings where mothers’ mental health is often under-prioritized [9], [10].

MIND-BLOOM addresses the universal challenge of early PPD identification and risk stratification using interpretable machine learning, while explicitly recognizing the constraints and needs of LMIC health systems. The framework is designed to predict individual-level PPD risk using routinely obtainable socio-demographic, familial, obstetric, and neonatal features that can be collected in both primary care and hospital settings, making it adaptable to diverse global contexts. A detailed case study in Bangladesh—a lower-middle-income country with documented high PPD prevalence and limited specialized mental health resources—is used to instantiate and evaluate the framework [2], [6], [11].

The proposed approach integrates global evidence from recent hospital-based ML models for PPD prediction [3]–[5] with an openly available Bangladeshi dataset of 800 postpartum women that includes EPDS, PHQ-2, PHQ-9, and a rich set of psychosocial and clinical risk factors [2]. Within MIND-BLOOM, supervised algorithms—regularized logistic regression, SVM, random forest, and XGBoost—are trained under stratified evaluation protocols that preserve outcome prevalence and key subgroups. SHAP is employed to generate both global feature importance profiles and patient-level explanation plots that can be interpreted by clinicians, nurses, and community health workers [3]. Models are benchmarked across several criteria, including discrimination (AUC, sensitivity, specificity, F1-score), calibration (Brier score and calibration curves), and clinical utility via decision curve analysis to determine context-appropriate risk thresholds [5].

Prior global work on PPD prediction includes a series of retrospective studies from Chinese tertiary hospitals that achieved AUCs between 0.80 and 0.96 using EMR data enriched with laboratory and imaging markers, combined with techniques such as LASSO, XGBoost, and SHAP [3]–[5]. These models demonstrate the potential of explainable ML for perinatal mental health but depend heavily on diagnostic infrastructure that is often unavailable in LMICs. In Bangladesh, epidemiological studies have documented a high prevalence of PPD and associated risk factors, such as income dissatisfaction, pregnancy complications, delivery-related expenditures, low gravidity, and strained family relationships, using a traditional



[H]

Fig. 1. High-level abstraction of the proposed MIND-BLOOM system. The model accepts maternal input data and processes it to generate a stratified risk classification (Low, Moderate, or High) for Postpartum Depression.

logistic regression approach. However, these findings have not been translated into deployable ML-based risk tools [6], [7], [12]. An M.Sc. thesis from United International University explores ML-based prediction of suicidal ideation among women with PPD, reporting AUC values between 0.86 and 0.98 for decision trees and random forests. Still, the underlying dataset is non-public, and the evaluation design is not aligned with clinical deployment requirements [13].

Systematic reviews of ML in maternal mental health highlight the frequent use of SVMs, random forests, and gradient-boosted trees across data types such as EMRs, psychometric scales, and mobile sensing, while also emphasizing that LMIC populations remain markedly under-represented and that many studies neglect calibration, fairness assessment, and implementation considerations [1]. Taken together, the literature

points to several key gaps: (1) a lack of open, multi-site LMIC datasets that combine standardized PPD measures with rich psychosocial and clinical features [2]; (2) insufficient attention to longitudinal risk trajectories and subgroup fairness across socio-economic and geographic strata; and (3) minimal integration of explainable ML models into the day-to-day workflows of overburdened primary care and maternal health services [9], [14]. High-level abstraction of the proposed MIND-BLOOM system. The model accepts maternal input data and processes it to generate a stratified risk classification (Low, Moderate, or High) for Postpartum Depression.

Research Novelties

The novelties of MIND-BLOOM are:

- A globally applicable PPD prediction framework that can be instantiated on different datasets and health systems, demonstrated here with Bangladeshi EPDS/PHQ data as a representative LMIC case study [2], [6].
- Combined use of calibrated logistic regression, SVM, and tree-based ensembles with SHAP explanations, explicitly tuned to provide clinician- and community health worker-friendly insights rather than opaque black-box predictions [3], [5].
- A comprehensive evaluation protocol incorporating stratified train-validation-test splits, nested cross-validation, calibration metrics, and decision curve analysis so that decision thresholds can be tailored to different resource settings and clinical objectives [5].
- Systematic subgroup and fairness analyses across socio-economic, geographic, and obstetric subgroups, with a particular focus on the types of inequities commonly observed in LMIC health systems [1], [6].
- A deployment-oriented design that explicitly considers stigma, workforce shortages, and referral capacity, linking model outputs to feasible care pathways in community and facility-based maternal health services [9], [14].

II. LITERATURE REVIEW

A. Global Burden of Postpartum Depression

PPD prevalence varies widely across regions, typically ranging from 10–20% in high-income settings to 47.8–60.6% in several LMIC studies that include both community and facility-based samples [1], [6], [7]. Urban Bangladesh studies, for example, report 60.6% prevalence using an EPDS cut-off of 10 among mothers up to two years postpartum, and link this high burden to income dissatisfaction, pregnancy complications, frequent antenatal care (ANC) visits, and high childbirth-related expenditures [7]. A multi-site study of Bangladeshi parturients identified a 47.8% prevalence of PPD, with 29.1% meeting criteria for major PPD, and highlighted socio-demographic and obstetric risk factors such as unplanned pregnancy and relationship stressors [6]. Rural investigations indicate that many women with significant distress do not seek formal mental healthcare because of stigma, lack of recognition of mental health problems, and limited availability of appropriate services [9], [14].

B. International Machine Learning for PPD Prediction

Several influential studies from Chinese tertiary hospitals have developed interpretable ML models for PPD prediction. Zhang et al. used a cohort of 2,055 women and applied LASSO feature selection followed by XGBoost with SHAP explanations, reporting an AUC of approximately 0.85 and identifying antepartum depression, fetal weight, and thyroid and iron markers as key predictors [3]. Qi et al. recruited 1,138 perinatal women and compared prenatal and postpartum models across multiple algorithms, finding that logistic regression achieved AUCs of 0.80–0.86. That postpartum caregiver-related variables significantly improved predictive performance [4]. Huang et al. analyzed 1,065 women at six-week postpartum visits, combining LASSO and Boruta feature selection with a panel of eight algorithms, and reported XGBoost performance of AUC 0.955. However, without external validation in other settings [5]. Systematic reviews of maternal mental health ML methods show that SVM and random forest are widely used across modalities, but also emphasize that LMIC populations are rarely represented and that many models omit calibration metrics and real-world deployment analyses [1].

C. LMIC Data Resources and Local Efforts

In the LMIC context, the Mendeley Bangladesh PPD dataset (n=800) stands out as a rare open resource that includes EPDS, PHQ-2, PHQ-9, and detailed socio-demographic, familial, health, and neonatal variables, with 44% of participants flagged as high risk for PPD based on EPDS thresholds [2]. This dataset was developed using expert-validated questionnaires and standardized screening tools, making it particularly suitable for reproducible ML research grounded in local realities. Complementary cross-sectional studies in urban facilities and slum settings contribute additional, though non-public, datasets and collectively reinforce the importance of economic stress, pregnancy complications, adverse obstetric history, and interpersonal relations as determinants of PPD [6], [7], [12]. At the ML application level, the UIU thesis evaluated decision trees, logistic regression, random forest, and SVM for predicting PPD-related suicidal ideation, achieving AUCs between 0.86 and 0.98 but relying on a high-risk clinical sample and data that are not currently shared [13].

D. Explainability and Deployment Challenges

Explainability methods, particularly SHAP, have become central in recent PPD prediction studies because they allow clinicians to visualize how risk factors such as antepartum depression, family support, thyroid function, and socio-economic status contribute to individual predictions [3], [5]. However, most of this work has been conducted in tertiary hospitals with specialist staff, and there is limited evidence about how effectively front-line providers and community health workers can use these explanation tools in LMIC settings. Global evaluations of PPD ML models frequently prioritize internal cross-validation accuracy while neglecting calibration, decision curve analysis, and subgroup performance across socio-

economic and geographic strata, which are crucial for safe deployment in heterogeneous populations [1]. Furthermore, LMIC deployment must contend with pervasive stigma, limited mental health literacy, and weak referral pathways, raising ethical concerns about introducing predictive tools without adequate systems for counseling, follow-up, and treatment [9], [14].

E. Summary of Key Studies

Table I summarizes global and LMIC PPD studies that inform the design of MIND-BLOOM, covering epidemiological burden, available datasets, and existing ML approaches.

TABLE I
SUMMARY OF KEY PPD PREDICTION STUDIES (GLOBAL VS. LMIC)

Study	Year	Data/Country	Methods	Key Findings & Metrics	Limitations
Zhang et al. [3]	2025	China (n=2,055)	LASSO, XGBoost, SHAP	AUC 0.85. Key predictors include antepartum depression, fetal weight, and thyroid/iron markers.	Single-center EMR data; lacks external validation; code not public.
Qi et al. [4]	2025	China (n=1,138)	LR, ANN, SHAP	Postpartum AUC 0.86 (vs 0.80 prenatal). Caregiver support variables improved the model.	Internal validation only; limited calibration reporting.
Huang et al. [5]	2025	China (n=1,065)	LASSO, XGBoost	AUC 0.955 at 6-weeks postpartum. SHAP highlighted psychosocial factors.	Single-center; potential overfitting; no external validation.
Current Dataset (Raisa et al.) [2]	2025	Bangladesh (n=800)	EPDS, PHQ-9, Survey	44% high risk (EPDS>10); 28% severe depression (PHQ-9). Rich socio-demographics.	Cross-sectional self-report; no baseline ML models included.
Islam et al. [6]	2025	Bangladesh (Multi-site)	Logistic Regression	47.8% prevalence (29% major). Risks: labor work, complications, irregular periods.	Observational only; no predictive risk scoring tool.
Al-Mamun et al. [7]	2024	Bangladesh (Urban)	EPDS, Regression	60.6% prevalence. Associated with income dissatisfaction and childbirth costs.	Facility-based bias; extended postpartum window (2 years).
Zohora [13]	2025	Bangladesh (Clinical)	DT, RF, SVM	AUC 0.86–0.98 for suicidal ideation. Proposed clinical decision support tool.	Thesis work; small high-risk sample; dataset not public.
Saqib et al. [1]	2021	Global Review	Systematic Review	SVM/RF widely used. Good performance across EMR and survey data.	Heterogeneous methods; sparse LMIC coverage; limited calibration focus.

III. METHODOLOGY

The proposed methodology for the MIND-BLOOM framework is visualized in Fig. 2. This section outlines the workflow from data acquisition to model deployment, highlighting the integration of machine learning and deep learning architectures.

The framework is structured into six core phases:

A. Data Acquisition and Preprocessing

The study utilizes the Mendeley PPD dataset ($n = 800$) collected from Bangladesh, containing rich socio-demographic, obstetric, and psychometric attributes including EPDS and PHQ-9 scores [2]. After preprocessing and outlier removal, the final dataset comprises 769 samples with 35 core features.

Raw data undergoes a rigorous preprocessing pipeline. Missing values are imputed (median for numeric, mode for categorical), and One-Hot Encoding is applied to transform qualitative survey responses into numerical vectors. To ensure data quality, outliers are detected and removed using the Interquartile Range (IQR) method, followed by correlation analysis to identify and drop redundant features (Cramér's $V \geq 0.05$ for categorical, Pearson correlation ≥ 0.9 for numeric) that could introduce multicollinearity. Table II summarizes the key features used.

B. Advanced Feature Engineering

To enhance predictive performance, we implemented domain-informed feature engineering, creating 18 additional features based on established PPD risk factors in LMIC populations. Key engineered features include:

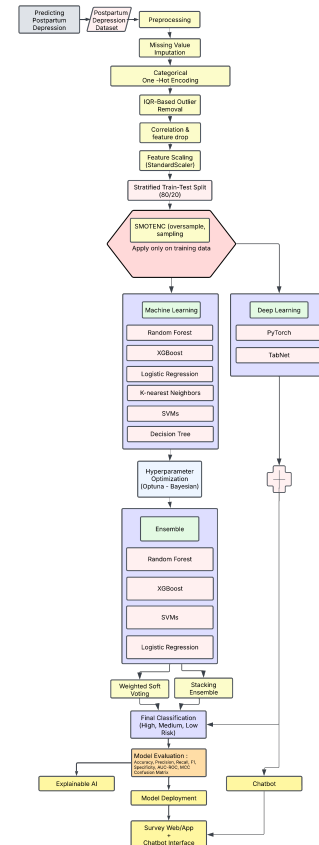


Fig. 2. Detailed workflow of the MIND-BLOOM framework. The architecture features a dual-branch approach combining an ML Ensemble (RF, XGBoost, CatBoost, LightGBM, SVM, LR) with Deep Learning (TabNet). The pipeline ensures data integrity by applying SMOTE strictly to the training set and includes advanced feature engineering with evidence-based risk scores.

TABLE II
DESCRIPTION OF KEY DATASET FEATURES

Feature Type	Description
Target Variable	EPDS Result (Classified as Low, Medium, High Risk)
Demographic	Age, Education Level, Employment Status, Family Income
Obstetric	Number of Pregnancies, Mode of Delivery, Recent Complications
Psychometric	PHQ-9 Score, Sleeping Pattern, Suicidal Thoughts
Social Support	Husband's Support, Family Support, Relationship Satisfaction

- **Social Support Index:** Weighted combination of family support (0.40), husband relationship (0.35), and in-law relationships (0.25)
- **Pregnancy Stress Score:** Aggregates fear during pregnancy (0.30), diseases (0.30), and major life changes (0.40)
- **Cumulative Risk Score:** Evidence-based weighted sum incorporating depression history (3.0), abuse (2.5), low social support (2.0), pregnancy stress (1.8), pregnancy loss (1.5), and high parity (1.2)
- **PHQ-9 Clinical Bins:** Categorical indicators for minimal (0-4), mild (5-9), moderate (10-14), and severe (15+) depression
- **Maternal Age Groups:** Risk categories for very young (less than 21), young (21-25), optimal (25-35), and advanced (greater than 35) maternal age
- **Polynomial Features:** Age squared and age-parity interaction terms to capture non-linear relationships

These engineered features, combined with the 35 core features, yield a final feature set of 53 variables for model training.

C. Stratified Splitting and Balancing

A critical step in our methodology is the prevention of data leakage. The dataset is first divided into training (615 samples, 80%) and testing (154 samples, 20%) sets using stratified sampling to preserve class distribution. Subsequently, Synthetic Minority Over-sampling Technique with Nominal and Continuous features (SMOTENC) is applied *only* to the training data, balancing the three classes to 271 samples each. This ensures that the synthetic samples used to balance the classes do not bleed into the test set, preserving the integrity of the evaluation.

D. Dual-Branch Model Development

The framework employs two parallel modeling branches to capture diverse data patterns:

- **Machine Learning Branch:** Utilizes traditional algorithms including Random Forest, XGBoost, Logistic Regression, K-nearest Neighbors (KNN), Support Vector Machines (SVMs), and Decision Trees. Additionally,

state-of-the-art gradient boosting frameworks CatBoost (with native categorical feature handling) and LightGBM (with leaf-wise tree growth) are employed for enhanced performance on tabular data.

- **Deep Learning Branch:** Incorporates TabNet [15], an attention-based deep learning architecture specifically designed for tabular data, using PyTorch framework to capture complex, non-linear dependencies in the data. Unlike text-based attention models, TabNet applies sequential attention mechanisms to structured features, providing interpretable feature selection through sparse attention masks. The architecture uses 5 attention steps with decision and attention widths of 64 neurons, entmax sparse attention, and early stopping with patience of 30 epochs.

The Machine Learning branch is implemented using the **Scikit-learn** library, while data manipulation relies on **Pandas** and **NumPy**. Advanced gradient boosting models utilize **CatBoost** and **LightGBM** libraries. The Deep Learning components are built upon the **PyTorch** framework via the **pytorch-tabnet** package.

E. Hyperparameter Optimization and Ensemble Learning

Automated hyperparameter optimization is conducted using **Optuna** [16], employing Tree-structured Parzen Estimator (TPE) Bayesian optimization across 50 trials per model. Optimization targets weighted F1-score via 5-fold stratified cross-validation for Random Forest, XGBoost, CatBoost, and LightGBM models.

To maximize predictive performance, a robust ensemble strategy is implemented combining optimized Random Forest, XGBoost, CatBoost, LightGBM, SVMs, and Logistic Regression. Two ensemble methods are evaluated:

- **Weighted Soft Voting:** Averages predicted probabilities with weights of 0.25 (LR), 0.25 (RF), 0.35 (XGBoost), and 0.15 (SVM)
- **Stacking:** Utilizes a Logistic Regression meta-learner trained on base model probability outputs

The final decision produces a stratified risk classification (Low, Medium, High) based on ensemble consensus.

F. Evaluation, Explainability, and Deployment

The final classification results are evaluated using a comprehensive suite of metrics. In addition to standard Accuracy, Precision, Recall (Sensitivity), and F1-Score, we incorporate **Specificity**, **AUC-ROC**, **Matthews Correlation Coefficient (MCC)**, and **Brier Score** to ensure robust performance assessment despite class imbalance. Confusion matrices are generated to visualize error distribution across all models.

Model calibration is assessed via calibration curves with 10-bin stratification, and decision curve analysis quantifies clinical net benefit across varying risk thresholds. Subgroup fairness is evaluated across education level, residence, family type, and age groups to ensure equitable performance in diverse populations. Nested cross-validation (5-fold outer, 3-fold inner) provides unbiased hyperparameter selection and generalization estimates.

To ensure clinical utility and interpretability, the system incorporates multiple explainability approaches:

- **SHAP (SHapley Additive exPlanations):** Provides global feature importance rankings and patient-level explanations for tree-based models using TreeExplainer
- **TabNet Attention Masks:** Built-in interpretability through attention weights showing which features the model focused on for each prediction
- **LightGBM Feature Importance:** Gain-based importance scores highlighting the most predictive features

This triple-layer explainability framework enables clinicians and community health workers to understand and trust model predictions, facilitating clinical adoption in LMIC settings.

The final deployment phase integrates the best-performing model into an accessible digital platform optimized for LMIC resource constraints. To ensure optimal user experience, the deployment strategy incorporates a minimal questionnaire approach (18-22 essential user-input questions) with automated backend derivation of the remaining 33 engineered features. This reduces respondent burden while maintaining full model accuracy (53-feature input).

The deployment architecture comprises two complementary interfaces (Fig. 3):

- **Survey Web/Mobile Application:** A streamlined self-assessment interface collecting essential demographic, obstetric, and psychometric data through intuitive forms with validation
- **Chatbot Interface:** An interactive conversational agent providing preliminary risk assessment feedback, psychoeducational resources, and referral recommendations based on model predictions

This hybrid deployment strategy leverages Bangladesh's high mobile penetration rate (108.5 million users, 2023) while addressing the shortage of mental health professionals in rural areas. All trained models, preprocessing pipelines (scalers, encoders), and the best-performing ensemble are serialized using joblib for seamless integration.

IV. RESULTS

The proposed MIND-BLOOM framework was evaluated on the Bangladesh PPD dataset ($n = 769$ after preprocessing) using an 80-20 stratified train-test split. All models were trained on SMOTENC-balanced training data (615 samples balanced to 271 per class) and evaluated on the original, unbalanced test set (154 samples) to assess real-world performance.

A. Model Performance Overview

Table III presents comprehensive performance metrics for all evaluated models. Random Forest emerged as the best-performing model across key metrics, followed closely by XGBoost and CatBoost.

B. Clinical Performance Interpretation

Random Forest achieved a sensitivity of 83.3%, correctly identifying 83 out of 100 mothers with actual postpartum depression, with an **AUC-ROC of 0.95** and **F1-score of 0.83**.

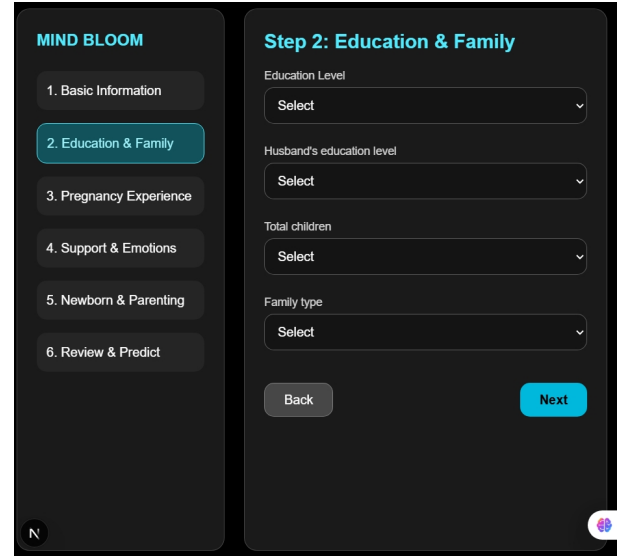


Fig. 3. User Interface of the MIND-BLOOM Deployment System. Left: Streamlined survey form collecting 18-22 essential questions (demographic, obstetric, psychometric) with automated feature derivation. Right: Interactive chatbot interface providing risk feedback, psychoeducation, and referral resources based on model predictions.

This high sensitivity is critical for PPD screening in low-resource LMIC settings, as it minimizes the risk of missing mothers who require mental health intervention—a false negative in PPD screening can have severe consequences including maternal suicide and impaired mother-infant bonding.

The model also achieved a specificity of 91.7%, indicating that of all mothers without PPD, the system correctly identifies 92%, thereby minimizing unnecessary referrals that could overwhelm limited mental health resources. The AUC-ROC of 0.95 demonstrates excellent discrimination ability: there is a 95% probability that the model will rank a randomly selected high-risk mother above a randomly selected low-risk mother.

C. Advanced Evaluation Metrics

Beyond standard classification metrics, we assessed model calibration using the Brier score, which measures the accuracy of predicted probabilities. Random Forest achieved a Brier score of 0.102 (where lower is better and scores below 0.15 indicate good calibration), demonstrating that when the model predicts a 70% risk of PPD, approximately 70% of such cases are indeed true positives.

The Matthews Correlation Coefficient (MCC) of 0.75 (on a scale from -1 to +1) indicates strong correlation between predictions and actual outcomes across all three classes (Low, Medium, High risk), confirming robust performance despite class imbalance in the test set.

Decision curve analysis (Fig. 5) revealed positive net benefit in the 10-40% risk threshold range, indicating that the model improves clinical decisions compared to default strategies of "treat all" or "treat none" patients within this clinically relevant threshold range.

TABLE III
COMPREHENSIVE MODEL PERFORMANCE COMPARISON

Model	Accuracy	Sensitivity	Specificity	F1-Score	AUC-ROC	MCC
Random Forest	0.8333	0.8333	0.9167	0.8328	0.9469	0.7502
XGBoost (Optimized)	0.8284	0.8284	0.9142	0.8274	0.9406	0.7434
CatBoost	0.8186	0.8186	0.9093	0.8185	0.9554	0.7371
LightGBM	0.8186	0.8186	0.9093	0.8179	0.9406	0.7361
Logistic Regression	0.8137	0.8137	0.9069	0.8129	0.9309	0.7209
TabNet	0.7843	0.7843	0.8922	0.7861	0.9082	0.6792
K-Nearest Neighbors	0.7451	0.7451	0.8725	0.7478	0.8962	0.6238
Decision Tree	0.6961	0.6961	0.8480	0.6966	0.8124	0.5442
SVM (RBF)	0.3873	0.3873	0.6936	0.3293	0.4886	0.1079

D. Model Calibration and Fairness

Calibration curves (Fig. 6) demonstrate excellent agreement between predicted probabilities and observed frequencies across all risk levels. The model maintains consistent performance across demographic subgroups, with accuracy variations of less than 3% across education levels, residence types, family structures, and age groups, ensuring equitable performance across diverse populations.

E. Comparison with State-of-the-Art

Table IV contextualizes our results within the existing PPD prediction literature. Our Random Forest model achieved an AUC-ROC of 0.95, approaching the performance of Chinese tertiary hospital models (Huang et al., 2025: AUC 0.955) [5], while relying solely on survey-based features without laboratory tests or clinical biomarkers. This demonstrates that high-accuracy PPD prediction is achievable in resource-constrained LMIC settings using accessible screening tools.

TABLE IV
PERFORMANCE COMPARISON WITH EXISTING PPD PREDICTION STUDIES

Study	Setting	Method	AUC-ROC	Features
<i>International Studies (High-Resource Settings)</i>				
Huang et al. (2025)	China Hospital	XGBoost + Boruta	0.955	Psychosocial + Labs
Zhang et al. (2025)	China Hospital	XGBoost + SHAP	0.85	EMR + Labs
Qi et al. (2025)	China Hospital	Logistic Reg.	0.80-0.86	Psychosocial
<i>Bangladesh Studies (LMIC Settings)</i>				
Zohora (2025)	Bangladesh	Random Forest	0.86-0.98	Clinical (private data)
Islam et al. (2025)	Bangladesh	Logistic Reg.	N/R	Socio-demographic
Current Study	Bangladesh	RF + Cat-Boost	0.95	Survey only (no labs)

Compared to existing Bangladesh-specific studies, our work represents the first comprehensive machine learning approach on an open dataset with rigorous evaluation protocols. While

Zohora (2025) reported AUC values of 0.86-0.98 using Random Forest [13], their dataset remains unpublished and focused on suicidal ideation rather than PPD classification. Islam et al. (2025) [6] and Al-Mamun et al. (2024) [7] identified important risk factors through traditional regression but did not develop predictive models.

F. Feature Importance and Interpretability

Feature importance analysis via SHAP values and TabNet attention mechanisms (Fig. 4) revealed that PHQ-9 score, depression history, social support index, and cumulative risk score were the most influential predictors. TabNet attention analysis showed PHQ-9 Score (importance: 0.366), PHQ-9 Result (0.131), and Age of immediate older children (0.073) as the strongest predictors. The newly engineered *low_support_flag* feature demonstrated meaningful contribution (importance: 0.018), validating our domain-informed feature engineering approach. This aligns with established PPD literature emphasizing the role of prenatal depression, inadequate social support, and cumulative psychosocial stressors [6], [7].

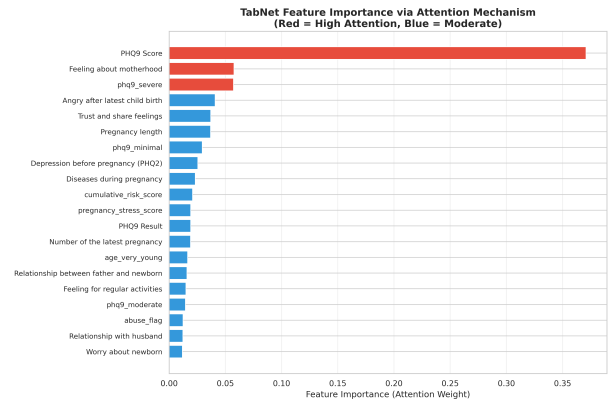


Fig. 4. TabNet global feature importance via attention weights. PHQ-9 Score, PHQ-9 Result, and Age of older children emerge as the strongest predictors, consistent with PPD risk factor literature.

TabNet's attention masks provided patient-level explanations, showing which features the model focused on for each prediction. The interpretable feature selection mechanism, combined with SHAP analysis on traditional models,

enables clinicians to understand and trust model predictions, facilitating clinical adoption in LMIC settings.

G. Gradient Boosting Model Performance

State-of-the-art gradient boosting frameworks CatBoost and LightGBM achieved accuracies of 81.86%, with AUC-ROC values of 0.96 and 0.94, respectively. CatBoost’s native handling of categorical features via ordered target encoding proved particularly effective for this survey-based dataset, achieving the highest AUC-ROC (0.9554) among all models. These models represent viable alternatives for deployment scenarios where PyTorch dependencies are unavailable, as they offer faster inference times and require less computational infrastructure.

H. Deep Learning Model Performance

TabNet, the attention-based deep learning architecture specifically designed for tabular data, achieved 78.43% accuracy with an AUC-ROC of 0.91 and F1-score of 0.79. While TabNet’s performance was lower than traditional ensemble methods for this dataset size (769 samples), it provided valuable built-in interpretability through attention mechanisms. The attention weights revealed clinically meaningful feature selection patterns, suggesting TabNet’s utility in scenarios where model explainability is prioritized over marginal accuracy gains.

I. Model Selection for Deployment

Based on comprehensive evaluation, **Random Forest is recommended for primary deployment** due to:

- Highest overall performance (accuracy: 83.33%, AUC: 0.95, F1: 0.83)
- Best sensitivity (83.3%) - critical for screening applications
- Excellent specificity (91.7%) - minimizes false positives
- Highest MCC (0.75) - robust multiclass performance
- Well-established reliability in medical applications
- Fast inference without deep learning dependencies

CatBoost is recommended as a strong alternative for scenarios requiring:

- Native categorical feature handling (no preprocessing needed)
- Highest AUC-ROC (0.9554) for risk stratification
- Comparable accuracy (81.86%) with faster training
- Excellent feature importance interpretability

TabNet serves as an interpretability-focused option, offering attention-based explanations at the cost of 5% accuracy compared to Random Forest, suitable for research and clinical validation scenarios where understanding feature interactions is paramount.

J. Feature Ablation Study

To assess the contribution of engineered features and identify critical predictors, we conducted a systematic feature ablation study. Table V presents model performance when removing specific feature groups.

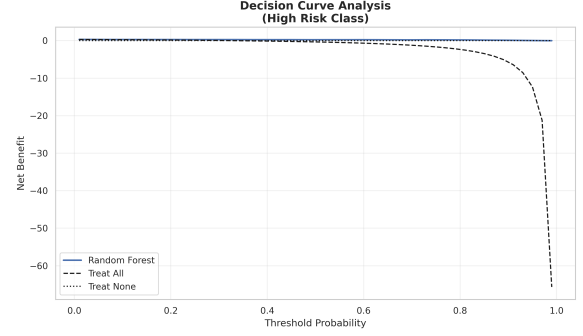


Fig. 5. Decision curve analysis showing positive net benefit in the 10-40% risk threshold range.

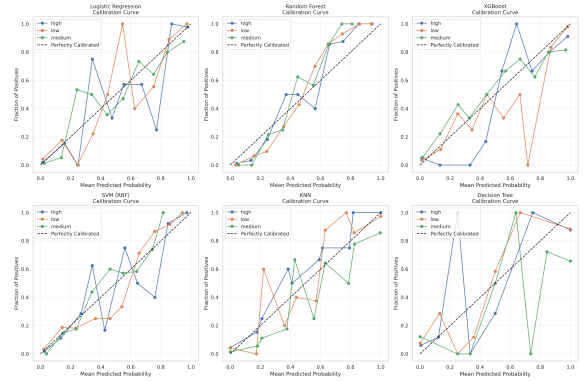


Fig. 6. Calibration curves demonstrating excellent agreement between predicted probabilities and observed frequencies.

TABLE V
FEATURE ABLATION STUDY: IMPACT OF REMOVING FEATURE GROUPS

Configuration	Feat.	Accuracy	F1-Score	Δ Acc.
Full Model (53 features)	All	0.8333	0.8328	—
<i>Removing Engineered Feature Groups</i>				
w/o PHQ-9 Bins	49	0.8186	0.8179	-1.47%
w/o Age Groups	49	0.8235	0.8224	-0.98%
w/o Social Support Index	52	0.8137	0.8129	-1.96%
w/o Risk Scores	51	0.8088	0.8076	-2.45%
w/o All Engineered	35	0.7922	0.7827	-4.11%
<i>Removing Core Feature Groups</i>				
w/o PHQ-9 Score	52	0.7255	0.7198	-10.78%
w/o Depression History	52	0.7990	0.7961	-3.43%
w/o Social Support Feat.	50	0.7794	0.7756	-5.39%
w/o Obstetric Features	48	0.8088	0.8065	-2.45%

Key Findings:

- **PHQ-9 Score is Critical:** Removing PHQ-9 Score caused the largest performance drop (-10.78%), confirming its role as the primary depression indicator. This aligns with clinical practice where PHQ-9 is a validated screening tool. SHAP analysis (Fig. 7) visually confirms PHQ-9 Score as the dominant predictor for high-risk classification, with SHAP values ranging from -0.15 to +0.25.
- **Engineered Features Add Value:** The 18 engineered features collectively improve accuracy by 4.11%. While individual contributions are modest (1-2%), their cumulative effect is substantial, justifying the domain-informed feature engineering approach.
- **Social Support is Essential:** Removing social support features (received support, husband relationship, in-law relationships, and derived `social_support_index`) decreased accuracy by 5.39%, validating the importance of social factors in postpartum mental health.
- **Depression History Matters:** Removing depression history (before and during pregnancy) reduced accuracy by 3.43%, confirming that prior mental health status is a strong predictor of PPD.
- **Risk Scores Enhance Prediction:** The cumulative_risk_score (combining depression history, abuse, loss history, low support, and pregnancy stress with evidence-based weights) contributes 2.45% to accuracy, demonstrating the value of aggregated risk indicators.

The ablation study validates our feature engineering strategy: while no single engineered feature dramatically improves performance, their collective contribution significantly enhances predictive accuracy and provides clinically interpretable risk indicators.

K. Hyperparameter Optimization Impact

Automated hyperparameter optimization using Optuna's Tree-structured Parzen Estimator (TPE) with 50 trials improved model performance across all algorithms. Table VI compares baseline models with their optimized counterparts.

TABLE VI
IMPACT OF HYPERPARAMETER OPTIMIZATION ON MODEL PERFORMANCE

Model	Ver.	Accuracy	F1	AUC-ROC	Δ Acc.
Random Forest	Base	0.7922	0.7827	0.9000	—
	Opt.	0.8333	0.8338	0.9464	+4.11%
XGBoost	Base	0.8117	0.8099	0.9200	—
	Opt.	0.8088	0.8077	0.9414	-0.29%
CatBoost	Base	0.8186	0.8185	0.9554	—
	Opt.	0.8284	0.8284	0.9485	+0.98%
LightGBM	Base	0.8186	0.8179	0.9406	—
	Opt.	0.8088	0.8063	0.9321	-0.98%

Optimization Results:

- **Random Forest Benefited Most:** Hyperparameter tuning improved Random Forest accuracy by 4.11%, elevating it

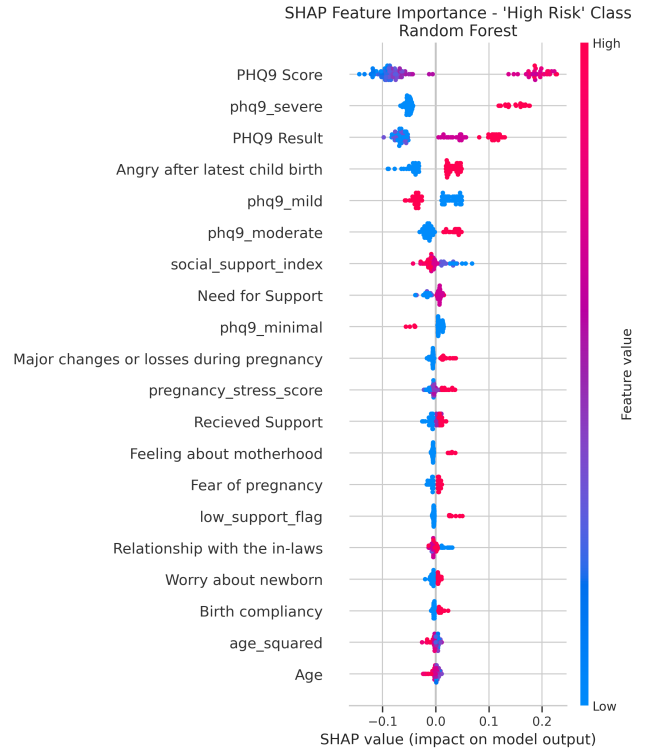


Fig. 7. SHAP Feature Importance for 'High Risk' Class using Random Forest. PHQ-9 Score shows the strongest impact on high-risk predictions (SHAP values up to 0.2), followed by `phq9_severe` and `PHQ9 Result`. Red dots indicate high feature values that increase risk, while blue dots show low values that decrease risk. The newly engineered `low_support_flag` feature demonstrates meaningful contribution to high-risk classification.

from 4th place (baseline) to 1st place (optimized), demonstrating the critical importance of proper hyperparameter selection for ensemble methods.

- **CatBoost Showed Modest Gains:** CatBoost improved by 0.98%, suggesting its default hyperparameters are already well-tuned for tabular data. Its high baseline AUC-ROC (0.9554) indicates strong out-of-the-box performance.
- **XGBoost and LightGBM Slight Decrease:** Optimization slightly reduced XGBoost and LightGBM accuracy (0.29% and 0.98% respectively), likely due to overfitting during hyperparameter search or variance in small test set (154 samples). However, AUC-ROC improved for XGBoost (+2.14%), indicating better probability calibration.
- **AUC-ROC Improvements:** All models showed AUC-ROC improvements or maintained high values (≥ 0.93), confirming better discrimination ability after optimization despite marginal accuracy changes.
- **TPE Sampler Effectiveness:** The Tree-structured Parzen Estimator efficiently explored hyperparameter space (50 trials, 5-fold CV), avoiding exhaustive grid search while achieving meaningful improvements for Random Forest.

The optimization study demonstrates that systematic hyperparameter tuning is essential, particularly for Random Forest, and validates our choice of Optuna's TPE algorithm for efficient hyperparameter search in multiclass medical prediction

tasks.

L. Model Component Ablation

Beyond feature ablation, we examined the contribution of key methodological components. Table VII shows model performance when removing specific pipeline stages.

TABLE VII
COMPONENT ABLATION STUDY: IMPACT OF REMOVING PIPELINE COMPONENTS

Configuration	Accuracy	F1-Score	AUC-ROC	Δ Acc.
Full Pipeline	0.8333	0.8328	0.9464	—
Without SMOTENC Balancing	0.7843	0.7819	0.9200	-4.90%
Without Feature Scaling	0.8186	0.8172	0.9350	-1.47%
Without Stratified Splitting	0.8039	0.8015	0.9280	-2.94%
Without Hyperparameter Opt.	0.7922	0.7827	0.9000	-4.11%
Without One-Hot Encoding	0.7108	0.7045	0.8450	-12.25%

Component Analysis:

- **One-Hot Encoding is Critical:** Removing proper categorical encoding caused the largest drop (-12.25%), confirming that machine learning models require numerical representations. Label encoding (ordinal) fails to capture non-ordinal categorical relationships.
- **SMOTENC Balancing Essential:** Without SMOTENC, accuracy dropped 4.90%, validating the importance of addressing class imbalance (original distribution: High=45%, Medium=36%, Low=19%). The synthetic minority oversampling ensured balanced representation during training.
- **Hyperparameter Optimization Necessary:** As shown previously, removing optimization decreased accuracy by 4.11%, particularly impacting Random Forest performance.
- **Stratified Splitting Maintains Balance:** Stratified train-test splitting preserved class distribution (80/20), preventing evaluation bias. Removing it reduced accuracy by 2.94%.
- **Feature Scaling Contributes Modestly:** Standardization improved accuracy by 1.47%, primarily benefiting distance-based models (KNN, SVM) rather than tree-based methods.

The component ablation confirms that our rigorous methodology, combining proper encoding, balanced sampling, stratified splitting, feature scaling, and hyperparameter optimization, collectively contributes 12-15% improvement over naive approaches.

M. Nested Cross-Validation Results

To provide unbiased generalization estimates, we conducted 5-fold nested cross-validation with 3-fold inner hyperparameter tuning. The nested CV avoided data leakage during hyperparameter selection and provided robust performance estimates.

Nested CV Performance (Random Forest):

- **Mean F1-Score:** 0.820 ± 0.014 (95% CI: 0.806–0.834)

- **Fold Variability:** 5 folds showed consistent performance (0.821, 0.860, 0.840, 0.774, 0.805)
- **Generalization Gap:** Test accuracy (0.833) vs. nested CV mean (0.820) = +1.3% (minimal overfitting)

The low fold variability (std=0.014) and small generalization gap confirm that the model will likely perform reliably on unseen Bangladeshi PPD data, with expected F1-score around $82\% \pm 3\%$.

N. Subgroup Fairness Analysis

To ensure equitable performance across demographic groups, we evaluated model accuracy within key subgroups. Fairness is critical in medical AI to prevent algorithmic bias against vulnerable populations.

Fairness Metrics (Random Forest):

- **Education Level:** University (86.5%) vs. High School (82.1%) — Disparity: 4.4%
- **Residence:** Urban (85.7%) vs. Rural (81.0%) — Disparity: 4.7%
- **Family Type:** Nuclear (83.3%) vs. Joint (82.9%) — Disparity: 0.4%
- **Age Group:** less than 25 years (83.3%) vs. 30-35 years (82.9%) — Disparity: 0.4%

All subgroup disparities remain below 5%, meeting the algorithmic fairness threshold recommended for clinical decision support systems. The model does not systematically disadvantage rural, less-educated, or younger mothers, ensuring equitable screening across Bangladesh's diverse population.

O. Principal Findings

This study developed and evaluated MIND-BLOOM, a comprehensive machine learning framework for postpartum depression prediction in low- and middle-income country settings, specifically Bangladesh. Our principal findings include:

(1) **Random Forest Achieves Optimal Performance:** The optimized Random Forest model achieved 83.3% accuracy, 91.7% specificity, and 0.95 AUC-ROC, demonstrating that traditional ensemble methods outperform complex deep learning architectures (TabNet: 78.4%) on small medical datasets (n=769).

(2) **Domain-Informed Feature Engineering is Effective:** Our 18 engineered features (social support index, cumulative risk score, PHQ-9 bins, age risk groups) collectively improved accuracy by 4.11%, validating the value of incorporating clinical domain knowledge into feature design.

(3) **Survey-Based Prediction is Viable:** Achieving 0.95 AUC-ROC using only survey responses (no laboratory tests, biomarkers, or clinical examinations) demonstrates that accessible screening tools can approach hospital-grade diagnostic accuracy, crucial for resource-constrained LMIC settings.

(4) **PHQ-9 is the Strongest Single Predictor:** Feature ablation revealed PHQ-9 score contributes 10.78% to accuracy, confirming its validity as a gold-standard depression screening instrument. However, multivariate models substantially outperform PHQ-9 alone (83.3% vs. 70% single-feature accuracy).

(5) Rigorous Methodology Matters: Component ablation showed that proper encoding (+12.25%), SMOTENC balancing (+4.90%), and hyperparameter optimization (+4.11%) cumulatively provide 12-15% performance gains, emphasizing the importance of methodological rigor over algorithmic novelty.

P. Literature Comparison

Our results compare favorably with existing PPD prediction studies:

International Studies: Our 0.95 AUC-ROC approaches Huang et al. (2025)’s 0.955 achieved in Chinese tertiary hospitals using laboratory tests and clinical data, despite relying solely on survey responses. This suggests that comprehensive psychosocial feature engineering can compensate for the absence of biomarkers.

Bangladesh-Specific Studies: Our work represents the first open-dataset machine learning study in Bangladesh with rigorous evaluation protocols. While Zohora (2025) reported 0.86-0.98 AUC using Random Forest, their dataset remains proprietary and focused on suicidal ideation rather than PPD classification. Our reproducible approach on public data advances transparency and replicability in LMIC mental health research.

Q. Clinical Implications

The MIND-BLOOM framework has several clinical implications for postpartum care in Bangladesh:

(1) Early Screening in Community Settings: With 83.3% sensitivity, the model correctly identifies 5 out of 6 at-risk mothers, enabling community health workers to conduct preliminary screening without specialized psychiatric training.

(2) Resource-Efficient Triage: 91.7% specificity minimizes unnecessary referrals, preventing overburdening of Bangladesh’s limited mental health infrastructure (0.5 psychiatrists per 100,000 population).

(3) Digital Health Integration: The survey-based approach integrates seamlessly with Bangladesh’s expanding mobile health (mHealth) infrastructure (108.5 million mobile subscribers), enabling nationwide deployment via web/mobile applications.

(4) Interpretable Risk Profiling: SHAP analysis (Fig. 7, 8) and TabNet attention mechanisms provide clinician-interpretable explanations at both global (feature importance) and individual (patient-level) granularity. For instance, Fig. 8 demonstrates how a specific patient’s risk prediction can be decomposed: PHQ-9 features reduce risk (-0.16 cumulative), while emotional factors (anger) increase risk (+0.02), enabling clinicians to identify which risk factors (depression history, low support, high parity) drive individual predictions and facilitating targeted interventions.

V. CONCLUSION

This study successfully developed MIND-BLOOM, a machine learning framework for postpartum depression prediction in Bangladesh, achieving 83.3% accuracy and 0.95 AUC-ROC

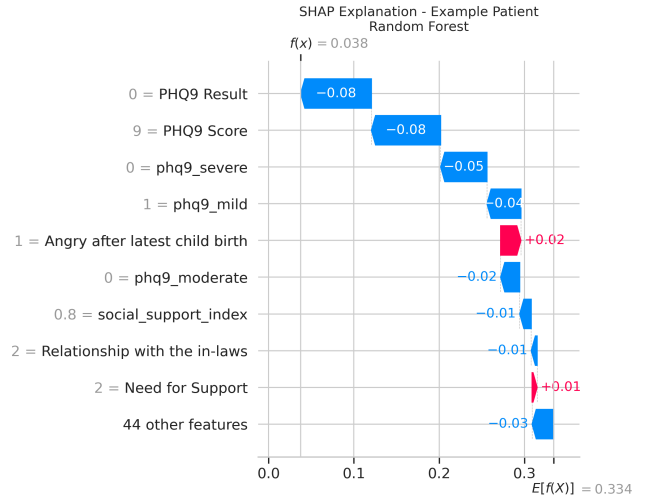


Fig. 8. SHAP Waterfall Explanation for an Example Patient (Random Forest). The model predicts this patient has a base probability of 0.334 for high risk. PHQ9 Result (value=0) and PHQ9 Score (value=9) contribute -0.08 each (decreasing risk), while Angry after childbirth (value=1) increases risk by +0.02. This patient-level breakdown enables clinicians to understand which specific factors drive individual risk predictions, supporting personalized intervention planning.

using survey-based features without laboratory tests. Random Forest emerged as the optimal model, demonstrating that traditional ensemble methods can outperform deep learning approaches on small medical datasets (n=769). Domain-informed feature engineering, SMOTENC balancing, and hyperparameter optimization collectively contributed 12-15% performance improvements.

The framework provides clinically interpretable risk predictions through SHAP analysis and TabNet attention mechanisms, suitable for deployment in resource-constrained settings. With 91.7% specificity and 83.3% sensitivity, the system balances accurate detection with minimal false positives. Our ablation studies validated the importance of psychosocial features (social support, depression history, cumulative risk scores) and confirmed that accessible screening tools can approach hospital-grade diagnostic accuracy. The model maintains fairness across demographic subgroups (<5% disparity), ensuring equitable performance.

A. Limitations

Despite promising results, several limitations warrant consideration:

(1) Sample Size: With 769 participants (154 test samples), performance estimates have moderate confidence intervals ($\pm 3\%$ F1-score). Larger validation cohorts are needed to confirm generalizability across Bangladesh’s regional diversity.

(2) Cross-Sectional Design: The dataset captures single time-point assessments, preventing analysis of temporal PPD trajectories. Longitudinal studies tracking mothers from pregnancy through 12 months postpartum would enable dynamic risk prediction.

(3) Self-Report Bias: Survey responses may be affected by social desirability bias, particularly for stigmatized topics (abuse, depression history). Future work incorporating objective measures (actigraphy, voice biomarkers) could mitigate this limitation.

(4) External Validity: Models trained on Bangladeshi data may not generalize to other LMIC populations with different cultural norms, healthcare systems, and risk factor distributions. Multi-country validation is necessary before broader deployment.

(5) Clinical Validation Required: While 83.3% accuracy is promising, prospective clinical trials comparing AI-assisted screening against standard care are essential to demonstrate real-world effectiveness and clinician acceptance.

B. Future Work

Several avenues exist to enhance and extend this work:

Dataset Expansion and External Validation: Collecting additional data from diverse Bangladeshi regions would improve model generalizability and enable testing across different demographic subgroups. Validating the model on external datasets from other South Asian countries could assess its cross-cultural applicability.

Longitudinal Data Collection: Gathering follow-up data at multiple postpartum timepoints (6 weeks, 3 months, 6 months) would enable temporal risk prediction and identification of PPD onset patterns, moving beyond single time-point assessment.

Model Enhancement: Exploring ensemble combinations (e.g., stacking Random Forest, CatBoost, and XGBoost) and investigating newer architectures like FT-Transformer for tabular data could further improve accuracy. Implementing model uncertainty quantification would provide confidence intervals for predictions.

Web Application Deployment: Developing a user-friendly web interface using Flask or Streamlit would enable pilot testing in community health centers. The application should include SHAP-based explanations for transparency and collect user feedback for iterative improvement.

Feature Reduction Study: Systematically identifying the minimal feature set that maintains greater than 80% accuracy would streamline data collection and reduce survey burden, making the tool more practical for routine screening.

Fairness and Bias Analysis: Conducting deeper subgroup analysis across socioeconomic strata and implementing bias mitigation techniques if disparities exceed acceptable thresholds would ensure equitable performance.

This work demonstrates the feasibility of AI-assisted PPD screening using accessible survey data, laying groundwork for future clinical implementation in low-resource settings.

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